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Volume 01 — M00–M11

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M00: Introductory Video — Information Security

Learning focus

- State why information security now spans individuals and enterprises.
- List the course's promised themes: web trust, DNS, HTTPS, client-side attacks, and secure coding.
- Explain the coordinator's claim that poor programming is the root cause of most information-security threats.
- Recall that this is a short, two-credit course on information and network security.

Key terms

Term	Meaning in this lecture
Security loophole	Organizational gap that lets an attacker break in and exploit a vulnerability
Authoritative DNS reply	Name-server answer that should be trusted when judging whether a site is genuine
HTTPS digital signature	Cryptographic check that the web server is the claimed site
DNS poisoning	Tampering with name resolution so users reach a fake host
Client-side attack	Compromise that succeeds because the user (browser) approaches a malicious site
Secure coding	The course's stated end-goal: stop shipping the vulnerabilities that attacks exploit

Exam lines

- Security is ambient: individual and enterprise alike.
- Trust on the web is technical (DNS + HTTPS), not visual.
- Poor programming, especially copy-paste reuse, is framed as the root cause of threats.
- Client-side attacks are hard to stop once the user cooperates with the attacker.
- The two-credit course aims at network, OS, database, DNS, and secure-coding competence.

M01: Information Security

Learning focus

- Define ethics and contrast consequentialism with deontology as approaches to right and wrong.
- Distinguish system security from information (data) security and state the CIA triad.
- Relate computer-security breaches and overly protective measures to rights, harms, and interests.
- Contrast hacking vs cracking, cyber trespass vs cyber vandalism, and cyber terrorism vs hacktivism vs information warfare.
- Explain why codes of ethics are not enough for information-security professionals.
- State why privacy matters, how IT enables dataveillance, and what concerns arise from public-space tracking and biometrics.

Key terms

Term	Meaning in this lecture
Ethics	Study of right/wrong and good/bad; analyzes behaviours, policies, laws, and social structures
Consequentialism	Wrongness of actions judged by bad consequences
Deontology	Moral duties exist independently of consequences
Computer ethics	1980s field on professional/user duties and IT public-policy ethics
System security	Protecting hardware/software against malicious sabotage
Information security	Protecting stored and transmitted data (CIA)
Safety-critical system	Real-time control system whose failure can threaten life
Hacking / cracking	Unauthorized access; crackers = malicious/disruptive break-ins (hacker self-description)
Hacker ethic	Information should be free; unlimited computer access; cyberspace cannot harm the real world
Cyber trespass	Unauthorized access to systems or password-protected sites
Cyber vandalism	Programs that disrupt networks or corrupt data

Term	Meaning in this lecture
Cyber terrorism	Politically motivated hacking intended to cause grave harm (death and/or severe economic loss)
Hacktivism	Political site disruption without intent to cause serious damage
Information warfare	Using information and attacks on information systems as tools of war
ISSA	Information Systems Security Association; code requires law plus "highest ethical principles" without specifying them
Right to be let alone	Warren and Brandeis's definition of privacy
Informed consent	Tell people how personal data will be stored/used/exchanged and obtain consent
Dataveillance	Large-scale computerized collection/processing of personal data to monitor actions and communications
Biometrics	Identity verification from physiological or behavioral characteristics

Exam lines

- Ethics is not the same as law: legality does not replace morality for individuals or corporations.
- Computer security has a moral structure: CIA plus system vs data, tied to rights, harms, and interests — including economic, cultural, psychological, and life-and-death stakes in safety-critical systems.
- Over-strong or discriminatory security can itself violate access, fairness, and freedom rights (including FOI and free speech when availability is blocked).
- The hacker ethic's "information should be free" claim collides with intellectual property, privacy, and data integrity.
- Cyber terrorism is marked by **political motive plus grave harm**; hacktivism and ordinary cyber crime are not the same thing; information warfare extends ordinary war onto information systems.
- Professional codes (e.g. ISSA) are too thin: security practitioners need ethics **training**, not only a code.
- Privacy is the right to be let alone / control access to private affairs; it is valuable for autonomy and democracy but not absolute.
- IT scaled surveillance into **dataveillance**; 9/11 and consumer tracking intensified the privacy–surveillance tension; public-space tracking and biometrics further squeeze anonymity.

M02: Network Security

Learning focus

- State why e-business networks need network security and list the CIA goals.
- Distinguish a **threat** from an **attack**, and **passive** from **active** attacks.
- Relate snooping, traffic analysis, replay, modification, and denial of service to confidentiality, integrity, and availability.

- Describe cryptography's three mechanisms: symmetric-key, asymmetric-key, and hashing (including MD5 and SHA).
- Place border routers, firewalls (static / stateful / proxy), IDS, and IPS in a network-boundary defense.
- List hallmarks of a good security policy and the non-technical side of defense in depth.

Key terms

Term	Meaning in this lecture
Confidentiality	Protect stored and transmitted information from unauthorized disclosure
Integrity	Authorized entities change data only through authorized mechanisms (faults can also violate it)
Availability	Authorized entities can access information when needed
Threat	Possible danger that might exploit a vulnerability
Attack	Deliberate assault that tries to evade services and violate policy
Passive attack	Eavesdropping / monitoring to obtain transmitted information
Traffic analysis	Inferring communication nature from patterns despite encryption
Snooping	Unauthorized access to or interception of data
Replay	Capture a data unit and retransmit it for an unauthorized effect
Symmetric-key crypto	Same shared secret key for encrypt and decrypt
Asymmetric-key crypto	Related public/private pair; decryption key infeasible to derive from encryption key
RSA	Example algorithm where either related key may encrypt and the other decrypt
Compression function	Fixed-size hash step: n bits in, m bits out, n > m
MD5	Rivest message digest: 512-bit blocks → 128-bit digest (too small)
SHA / NIST	Secure Hash Algorithm family standardized by NIST
ACL	Access control list on a border router (static L3 filter)
Ingress / egress filtering	Only properly destined traffic enters / leaves
Smurf attack	Spoofed ICMP echo to broadcast addresses; responses flood the victim
Stateful firewall	Tracks connections in a state table; most common type
Proxy firewall	Intermediary; full-packet protocol compliance; most advanced, least common
IDS	Detect and alert (signatures / anomaly); does not auto-block
IPS	Detect and automatically prevent/block
Defense in depth	Layered technical controls plus policy and awareness

Exam lines

- E-business needs scalable voice/video/data networks; more applications and users mean more threats, so network security is central.

- CIA is the goal set: conceal in storage and transit; authorize changes (including against accidental corruption); keep authorized access working (a bank that cannot serve accounts fails availability).
- A threat is potential danger; an attack is a deliberate intelligent assault. Passive attacks steal or infer information; replay, modification, and DoS produce unauthorized effects; snooping is stopped by making data meaningless via encryption.
- Cryptography today is three mechanisms: shared-key ciphers, public/private-key (RSA-style) algorithms, and iterated hashes (MD2/4/5; SHA from NIST). MD5's 128-bit digest is too small.
- Boundary defense: border router ACLs (ICMP, source routing, SYN, Smurf-style spoofing via ingress/egress), then firewalls (static vs stateful vs proxy), then IDS alerts and IPS auto-blocks.
- Defense in depth also needs a clear, scoped, dated, specific, understandable policy with named authority — not only tuned rule bases.

M03: Computer Network Reference Models

Learning focus

- Define a computer network and distinguish it from a distributed system (middleware / World Wide Web example).
- Classify networks by coverage: PAN, LAN, MAN, WAN, with named technologies and topologies.
- State the seven OSI-layer design principles and the PDU names from bit through APDU.
- Contrast **chained** layers (physical–network) with **end-to-end** layers (transport–application).
- List the function of each OSI layer (including MAC as a data-link sublayer).
- Describe the four-layer TCP/IP model, its ARPANET / DoD origins and design goals, key protocols, and how it differs from OSI.

Key terms

Term	Meaning in this lecture
Computer network	Autonomous computers interconnected by a single technology, able to exchange information
Distributed system	Independent computers appearing as one coherent system (often via middleware)
Middleware	Software above the OS that implements the distributed-system illusion
PAN / Bluetooth	~10 m personal interconnection; short-range radio
IEEE 802.3 Ethernet	Bus-based broadcast LAN, decentralized control, 10 Mbps–10 Gbps
IEEE 802.5 Token Ring	Ring LAN at 4 and 16 Mbps; transmit after capturing a token
FDDI	Another ring-network example

Term	Meaning in this lecture
WiMAX	Wireless MAN technology
Head-end	Central MAN point that feeds TV and Internet to homes
Subnet	WAN communication lines + switching elements/routers, typically telco/ISP owned
OSI	Open Systems Interconnection; ISO seven-layer model
Chained layers	Physical, data link, network — hop-by-hop, present on intermediate devices
End-to-end layers	Transport through application — only on end computers
TPDU / SPDU / PPDU / APDU	PDUs at transport, session, presentation, application
MAC	Medium Access Control sublayer: shared-channel access on broadcast networks
ARPANET	DoD-sponsored 1960s research network; grandfather of the Internet
TCP vs UDP	Reliable connection-oriented vs unreliable connectionless transport

Exam lines

- A network interconnects autonomous computers; a distributed system (WWW via middleware) makes them look like one machine.
- Coverage taxonomy: PAN (~10 m, Bluetooth), LAN (Ethernet bus / Token Ring / FDDI), MAN (cable TV, WiMAX, head-end), WAN (customer hosts plus telco/ISP subnet of lines and routers).
- OSI's seven layers follow explicit design principles; PDUs run bit → frame → packet → TPDU → SPDU → PPDU → APDU; lower three layers are chained, upper four are end-to-end.
- Each OSI layer has a distinct job: bits and media; frames, ACK, flow control, MAC; routing, congestion, addressing, QoS; end-to-end TPDUs and service type; sessions (dialog, tokens, sync); syntax/semantics; user protocols (HTTP, mail, news, file transfer).
- TCP/IP is the living Internet model: four layers, ARPANET/DoD goals (internetworking, surviving as long as endpoints live, flexibility), IP plus TCP/UDP plus Telnet/FTP/SMTP/DNS/NNTP/HTTP.
- Shared idea of a protocol stack and transport-to-applications split; OSI has more layers and a cleaner service/interface/protocol split; TCP/IP puts connectionless IP below and lets the user choose TCP or UDP above.

M04: Computer Network Reference Models — TCP/IP Layers Continuation

Learning focus

- Trace TCP/IP from ARPANET / DARPA (Kahn and Cerf, NCP) to the IETF-maintained Internet protocol suite.
- Map the four TCP/IP layers (and the five-layer hybrid that adds physical) onto OSI, including chained vs end-to-end layers.

- List link-layer tasks (MAC, LLC) and named link protocols: IEEE 802 family, ATM, ARP, NDP, LLTD, PPP.
- State IP's three functions and the internet-layer protocols IP, ICMP, IPsec/IKE, IGMP, and OSPF.
- Contrast TCP and UDP (RFCs, reliability, sockets/ports) and name DCCP, RSVP, TLS/SSL, and SCTP.
- Identify application-layer protocols with their RFCs: Telnet, FTP, SMTP, HTTP, RPC, DNS, SNMP, RTP.

Key terms

Term	Meaning in this lecture
Internet protocol suite / TCP/IP	Model and protocols of the Internet; named for TCP and IP
DoD model	Alternate name because ARPANET was DoD-sponsored
IETF	Body that maintains TCP/IP and related models
DARPA	Agency whose late-1960s R&D produced the suite
NCP	ARPANET Network Control Program; Cerf was its developer
Five-layer hybrid	Four TCP/IP layers plus an explicit physical layer 1
MAC / LLC	Shared-medium access vs flow control, link management, error handling, packet sizing
ATM	Fixed-size cells; mix real-time audio/video with regular data
ARP / NDP	IP→MAC mapping for IPv4 vs IPv6 neighbor discovery
LLTD	Microsoft proprietary topology / QoS discovery
PPP	ISP dial-up over a telephone line between two nodes
IP	Unreliable datagram forwarding; fragmentation; IPv4 and IPv6
ICMP	Errors and diagnostics (echo) sitting on IP
IPsec / IKE	Authenticate/encrypt IP datagrams; key exchange
IGMP	IP multicast group administration
OSPF	Link-state IGP inside one AS; called the most widely used Internet routing protocol here
Socket	IP address + port = unique connection endpoint
RFC 793 / RFC 768	TCP / UDP standards
DCCP / RSVP / SCTP	Congestion-aware datagrams; resource reservation; scalable stream TCP-like transport
TLS / SSL	Encrypt TCP segments; keys, auth, encryption, signatures

Exam lines

- TCP/IP is a practice-first, IETF-maintained DoD/ARPANET descendant: Kahn and Cerf hid network differences behind a common internet protocol and moved reliability from the network to the **host** (replacing NCP).
- Four layers (or five if physical is drawn separately): link and internet are chained; transport and application are end-to-end process communication. OSI protocols were designed first and are unused; TCP/IP dominates.

- Link layer is local (no routers): IEEE 802 (Ethernet, Token Ring, 802.11), ATM cells, ARP (IPv4) vs NDP (IPv6), Microsoft LLTD, and ISP PPP dial-up.
- Internet layer: hierarchical addressing, hop-by-hop routing, congestion control — IP (unreliable, fragmenting, v4/v6), ICMP, IPsec+IKE, IGMP multicast, OSPF inside an AS.
- Transport: ports and sockets; TCP (RFC 793, reliable bidirectional) vs UDP (RFC 768, fast, lossy, unordered); also DCCP, RSVP, TLS/SSL, SCTP.
- Application is the OSI 5–7 interface: Telnet, FTP, SMTP, HTTP, RPC, DNS, SNMP, RTP, several explicitly tied to RFCs and to client–server over TCP.

M05: Protocol Stack

Learning focus

- Distinguish a protocol **suite** (definition) from a protocol **stack** (software implementation) and state why layered modules exist.
- Recite the five OSI layering principles (including ISO 7498) and describe encapsulation/decapsulation (headers, Layer-2 trailer, physical signals).
- List services of each of the seven OSI layers, with named mechanisms (baud rate, line codes, CSMA/CD, IP/ICMP/IGMP/IPsec, TCP vs UDP, session beans, etc.).
- Describe the four TCP/IP layers and early applications Telnet, FTP, SMTP, and DNS (including the Yahoo name-to-address example).
- State the lecture's listed merits and demerits of TCP/IP.

Key terms

Term	Meaning in this lecture
Protocol suite	Definition of the protocols
Protocol stack	Software implementation of the suite; hierarchy of layers
ISO 7498	Identifier of the OSI model standard
D7…D1	Data units at OSI layers 7 down to 1
Baud / symbol rate	Symbol (or pulse) changes per second on the medium
Modem	Device that both modulates and demodulates
Line encoding	Unipolar, polar, bipolar, Manchester patterns of voltage/current/photons
Simplex / half / full duplex	One-way; either direction but not both at once; both at once
Frame	DLL PDU: link header + packet; trailer typically only here
ARQ	Retransmit specified frames when errors are detected
IEEE 802.2 LLC	Common LAN logical-link protocol
CSMA/CD vs token passing	Ethernet MAC vs Token Ring MAC
Slow start	Congestion avoidance that starts (or restarts after loss) at low rate
UDP-Lite	Lightweight UDP variant; still connectionless, no inherent congestion control

Term	Meaning in this lecture
Session beans	Session-scoped objects deleted when the web session disconnects
Network virtual terminal	Emulated terminal so a remote host accepts a logon
www.yahoo.com → 204.71.177.71	DNS example given in the lecture

Exam lines

- A suite is the specification; a stack is the running layered software, often split into media / transport / applications with two OS interfaces (e.g. TCP/IP ↔ Ethernet, browser ↔ TCP/IP).
- OSI (ISO 7498) uses seven abstraction layers with explicit design rules; data grows headers (and a layer-2 trailer) going down, becomes an EM signal, and is unwrapped going up.
- Physical: bits, baud, connectors, radio/IR, modulation/modems, line codes, bit sync, point-to-point vs multipoint, topology, duplex modes.
- Data link: frames, hardware addresses, flow control, ARQ, LLC (802.2), MAC (CSMA/CD vs token passing).
- Network: logical IP addresses, routing, datagrams, fragmentation, ICMP/IGMP/IPsec/IPv4/IPv6
- Transport PDUs traverse the network unmodified; TCP is a reliable connection; UDP/UDP-Lite are cheap and connectionless — applications must not blast at line rate. Ports multiplex (OSI would put that in session).
- Session (auth, authorization, restoration, checkpointing; session beans, lip-sync, live TV), presentation (translate, encrypt, compress), application (mail, virtual terminal, directories, FTAM).
- TCP/IP: underspecified host-to-network, connectionless IP, TCP or UDP (full or half duplex), Telnet/FTP/SMTP/DNS on top. Merits: independence, scale, client–server, many routing protocols, can connect two computers. Demerits: transport does not always guarantee delivery, poor reuse in other applications, hard protocol replacement, fuzzy service/interface/protocol split.

M06: Transmission Control Protocol

Learning focus

- Place TCP in the Cerf–Kahn 1974 internetworking paper and in RFC 793 / 1122 / 1323.
- Explain reliable stream delivery: segments, acknowledgements, timers, retransmission, and nesting TPDU ⊂ packet ⊂ frame.
- Define sockets, TSAPs/ports (including well-known ports below 1024 and IANA), and TCP's full-duplex, point-to-point, byte-stream service.
- Walk the 20-byte TCP header field by field (ports, seq/ack, data offset, flags, window, checksum, urgent pointer, options).
- Use socket primitives and the three-way handshake (plus simultaneous-open collision and FIN release / two-army timers).

- Apply variable-size sliding-window flow control, including a zero window and the one-byte probe that prevents deadlock.

Key terms

Term	Meaning in this lecture
Transmission control program (1974)	Cerf–Kahn component later split into TCP + IP
RFC 793 / 1122 / 1323	TCP definition; clarifications and bug fixes; extensions
Segment	TCP unit of transmission; IP encapsulates it for delivery
TPDU	Transport protocol data unit nested in packets then frames
Socket / TSAP	IP + 16-bit port; port is TCP's transport service access point
Well-known ports	Numbers below 1024 reserved for standard services; IANA assigns
Byte stream	Message boundaries not preserved; writes may coalesce or split
20-byte header	Fixed TCP header; options extra; max data 65495 with 20-byte IP
ACK number	Next byte expected, not last byte received
SYN/ACK/FIN/RST/PSH/URG	Connection setup, valid ACK, close, reset, push, urgent offset
Sliding window	Variable-size; WIN says how many bytes may be sent from ACK point
Pseudo-header	Included in the TCP checksum along with header and data
Three-way handshake	SYN; SYN+ACK; ACK (seq X / Y, ACK X+1 / Y+1)
Two-army problem	Connection-release uncertainty; solved with 2× MPL timers
Zero window	Sender stops except urgent data or a 1-byte window probe

Exam lines

- TCP grew out of the 1974 Cerf–Kahn transmission control program, then split from IP; it is specified in RFC 793 with later fixes (RFC 1122) and extensions (RFC 1323).
- It is a reliable, ordered **byte stream** with ACKs, retransmission timers, and segments carried inside IP packets inside frames — not a message stream, and not multicast/broadcast.
- Endpoints are sockets (IP + port/TSAP); well-known ports below 1024 (FTP 20, Telnet 23, SMTP 25/587, DNS 53, HTTP 80, POP/IMAP, ISAKMP 500, RTSP 554, etc.) are IANA-assigned.
- The 20-byte header numbers every byte, advertises a window, checksums a pseudo-header, and uses six flags for urgent data, valid ACKs, push, reset, setup (SYN), and teardown (FIN).
- Servers socket/bind/listen/accept; clients socket/connect; close is symmetric. Setup is a three-way handshake (RST if nobody listens); simultaneous open can collide. Teardown is two FINs (often four segments, sometimes three) plus timers against the two-army problem.
- Flow control is a variable window independent of data-link-style ACK coupling: a full 4K buffer can advertise WIN=0; urgent data and a one-byte probe still get through so a lost window update cannot deadlock.

M07: Data Link Layer

Learning focus

- Place the data link layer (DLL / Layer 2) above the physical layer and split it into LLC and MAC.
- List DLL functions: framing, hardware addressing, error control, flow control, and multi-access (CSMA/CD).
- Describe four framing methods: byte count, flag bytes with byte stuffing, flag bits with bit stuffing (01111110), and physical-layer coding violations (Manchester / IEEE 802).
- Contrast stop-and-wait with sliding-window flow control, including utilization U = 1/(1+2a).
- Explain ARQ error control: stop-and-wait ARQ, Go-Back-N ARQ, and selective-repeat ARQ.

Key terms

Term	Meaning in this lecture
DLL / Layer 2	OSI layer above physical; hides hardware; frames for upper layers
Collision domain	Shared-link setting that makes DLL work harder
LLC	Point-to-point sublayer: protocols, flow control, error control
MAC	Who uses the broadcast medium; CSMA/CD for shared access
NIC	Network interface card / adapter (e.g. Ethernet) implementing DLL in hardware
Segmentation / encapsulation	Split network-layer stream; add header and frame flags
Byte stuffing	Insert ESC before accidental flag bytes in the data
Bit stuffing	After five 1s in data, insert 0; flag is 01111110
Manchester / coding violation	1→10, 0→01; illegal physical codes delimit IEEE 802 frames
Stop-and-wait	One data frame, then wait for ACK; U = 1/(1+2a)
a	Propagation time / transmission time
Sliding window	Numbered frames; window up to 2^K−1; used by TCP (kernel buffer)
ARQ	Automatic repeat request after error detection
Stop-and-wait ARQ	ACK/NAK plus timer; duplicates discarded; small buffers, poor on long delay
Go-Back-N	Continuous send; on NAK/timeout resend that frame and all later ones; max window 2^K−1
RR	Receive Ready piggyback (e.g. RR1)
Selective repeat	Retransmit only lost/NAK frames; receiver resequences from storage

Exam lines

- Layer 2 sits on the physical bit pipe, hides the hardware, and is hardest on a shared collision domain; LLC (point-to-point protocols, flow, errors) and MAC (who speaks on the broadcast medium) split the work, often in an Ethernet NIC.
- Packets become frames with address, checksum, and control fields; four ways to mark frame edges are count, byte-stuffed flags, bit-stuffed 01111110, and Manchester coding violations (IEEE 802).
- Speed mismatches (Pentium vs 80386, Ethernet vs a 28.8 kbps modem) require flow control: stop-and-wait is simple but idle on long-delay paths (U = 1/(1+2a), bad for satellite); sliding windows of size at most 2^K−1 (and TCP's kernel buffer) keep multiple frames in flight and can cumulative-ACK.
- Errors are recovered by ARQ: stop-and-wait ARQ (ACK/NAK, timer, drop duplicates) is simplest and least efficient; Go-Back-N is continuous and rewinds to the bad frame (window capped at 2^K−1 so piggybacked RR cannot be ambiguous); selective repeat is most efficient and most complex because only lost frames are resent and the receiver must reorder.

M08: Medium Access Control Protocols

Learning focus

- Place the MAC sublayer in the OSI data-link layer and list its basic functions.
- Contrast static channel allocation (FDM) with dynamic allocation and state the LAN assumptions that dynamic methods need.
- Distinguish random-access (ALOHA, CSMA, CSMA/CD) from controlled-access (polling, token passing, slotted ring) methods.
- Compute Pure ALOHA and slotted ALOHA throughput and explain the vulnerable period in each.
- Describe nonpersistent, 1-persistent, and p-persistent CSMA, and the CSMA/CD collision procedure used on early Ethernet.

Key terms

Term	Meaning in this lecture
MAC sublayer	Bottom of the data-link layer; IEEE media-access, error detection, station addressing; serves LLC, uses the physical layer
LLC	Logical Link Control; MAC's user above
Multiple access	Many stations share one broadcast medium
FDM	Static split of bandwidth into fixed frequency partitions
Collision	Overlapping transmissions that garble the signal
Carrier sensing	Electrical sense that the channel is busy or free
ALOHA	Random-access protocol from the University of Hawaii (early 1970s)
Vulnerable period	Time window in which another packet can overlap and destroy a transmission
G, S	Offered load per packet time; successful throughput per slot
CSMA	Sense-before-transmit multiple access

Term	Meaning in this lecture
1-persistent CSMA	Keep sensing a busy medium, then send with probability 1 (Ethernet CSMA/CD)
p-persistent CSMA	After idle, send with probability p (CSMA/CA, Wi-Fi)
CSMA/CD	CSMA plus collision detection, jam, and backoff (early Ethernet)
10BASE5 / 10BASE2 / 10BROAD36 / 1BASE5	CSMA/CD cabling options named by Mbps, baseband/broadband, and segment length × 100 m

Exam lines

- Shared-medium LANs need MAC so simultaneous sends do not garble frames; MAC sits under LLC and above the physical layer.
- Static FDM wastes spectrum or starves users when traffic is not small and constant; dynamic methods are what LANs and WANs actually use.
- Random access (ALOHA, CSMA, CSMA/CD) versus controlled access (polling vs token/slotted ring) is the main protocol split; register insertion is obsolete.
- Pure ALOHA needs no sync but tops out near 18% ($S = Ge^{-2G}$ at $G = 0.5$); slotted ALOHA halves the vulnerable period and reaches about 36.8% ($S = Ge^{-G}$ at $G = 1$).
- CSMA listens first; 1-persistent is Ethernet/CSMA/CD, p-persistent is Wi-Fi/CSMA/CA.
- CSMA/CD aborts into a jam and backoff, and is taught as more efficient than ALOHA and plain CSMA on a bus with bursty traffic.

M09: TCP/IP Model-IP Addressing – I

Learning focus

- Trace a payload vertically down the TCP/IP stack from application to physical layer.
- State what a protocol is and when TCP versus UDP is used at the transport layer.
- Bind well-known applications to port numbers and describe a TCP segment's source and destination ports.
- Explain IPv4 dotted-decimal addressing, classful classes A–E, default masks, and network versus host bits.
- Compute network, broadcast, and gateway addresses, and state why CIDR/subnetting is introduced to cut classful waste.

Key terms

Term	Meaning in this lecture
All People Seem To Need Data Processing	Mnemonic for the seven layers from application down
Payload	Application data handed down (e.g. `www.google.com`)
Protocol	Set of rules for a line of action (TCP or UDP at transport)
Segment	Transport PDU: payload plus source and destination ports
Port	16-bit application identifier; 65,536 possible values
Well-known ports	Below 1024; not used as random source ports
Dotted-decimal	IPv4 written as four octets 0–255
Subnet mask	Marks which IP bits are network vs host
Unicast	One machine talking to one other machine
Network address	IP AND mask; host bits zero
Broadcast address	All host bits one
Gateway	Router address used to reach another network
CIDR	Classless inter-domain routing; subnetting to reduce classful waste

Exam lines

- Layers talk only **down the stack**; the browser payload becomes a TCP segment (ports) then an IPv4 packet (addresses).
- TCP is taught as the default for HTTP and mail; UDP is the other transport choice.
- HTTP listens on **80**; sources pick a port **above 1024**; FTP 20/21, SMTP 25, SSH 22, Telnet 23.
- IPv4 is 32-bit dotted decimal; classful A/B/C (with D multicast, E reserved) waste space when a site needs only 20 hosts.
- Network = IP AND mask; broadcast = host bits all 1; a gateway (often last host address) is required to leave the network; usable hosts are $2^n - 2$.
- CIDR/subnetting uses only as many host bits as needed (five bits for 20–31 hosts); mask arithmetic and MAC addresses are the next lecture.

M10: TCP/IP Model-IP Addressing – II

Learning focus

- Finish the 20-host CIDR example: mask `255.255.255.224`, block size, network and broadcast.
- Repeat the method for 10 hosts (`/28`) and list network, broadcast, first/last host, and wasted addresses.
- Write the same prefix in subnet-mask form and in CIDR slash notation.
- Distinguish IPv4 **logical** addresses from 48-bit **MAC physical** addresses (OUI + serial).
- Use ARP (broadcast request, unicast reply) to fill a frame when two hosts are on the same network; note that off-net frames use the **gateway's** MAC.

Key terms

Term	Meaning in this lecture
CIDR	Classless inter-domain routing; slash prefix equivalent to a mask
Incomplete octet	The mask octet that is not 0 or 255; used to get block size as $256-$ that value
Block size	Number of addresses in the CIDR block (e.g. 32 for mask 224, 16 for 240)
`/28`	28 network bits = mask `255.255.255.240`
Frame	Data-link PDU with source and destination MAC
Logical vs physical	IP vs MAC (phone number vs IMEI analogy)
OEM / OUI	First three MAC bytes: manufacturer
`ipconfig /all`	Shows IP, mask, and MAC
`ping`	Reveals the current IP of a name such as `www.google.com`
ARP	Broadcast request / unicast reply to map IP → MAC on the same network
Gateway MAC	Destination MAC of an off-net frame (not the remote server's MAC)

Exam lines

- For 20 hosts, mask **`255.255.255.224`**, block **32**, network `.0`, broadcast `.31`; leftover `.21`–`.30` are waste.
- For 10 hosts, **`192.168.1.0/28`** = **`255.255.255.240`**, block **16**, broadcast `.15`, 14 usable, four leftover if only 10 are assigned.
- Network layer = host-to-host **packet**; data-link = hardware **frame** with 48-bit MACs (24-bit OEM + 24-bit serial).
- Names become IPs (`ping`); IPs become MACs (`ipconfig /all` plus ARP).
- Same-net destination MAC comes from ARP; **off-net destination MAC is the gateway**, not the server — the usual student error.
- Transport = process-to-process (ports), network = host-to-host (IP), data-link = node-to-node (MAC).

M11: TCP Connection Management and Flow Control

Learning focus

- Explain how TCP can be connection-oriented while using connectionless IP.
- Walk the three-way handshake (passive/active open, SYN, SYN+ACK, ACK) including sequence-number consumption.
- Describe SYN flooding as a denial-of-service attack and the mitigations taught (limits, filters, cookies).
- Account for bidirectional data transfer, piggybacked ACKs, PSH, and URG (including what urgent mode is not).
- Contrast three-way connection termination with four-way half-close.

Key terms

Term	Meaning in this lecture
Virtual path	TCP's connection-oriented path over connectionless IP
Passive open	Server ready to accept; does not itself initiate
Active open	Client request to connect to a particular server
SYN	Control segment that syncs ISN; consumes one sequence number; no real data
SYN+ACK	Server's reverse SYN plus ACK of client SYN; carries rwnd
ACK (handshake 3)	Usually no data; consumes no sequence number
ISN	Random initial sequence number
TCB	Transmission control block; state allocated per embryonic connection
SYN flooding	DoS via forged SYNs that exhaust server resources
Cookie	Postpone allocation until the client IP is shown valid
Piggybacking	ACK carried with data
PSH	Push: send/deliver without waiting to fill buffers
URG / urgent pointer	Mark a stream range for special treatment; not priority or out-of-order delivery
FIN	Close in one direction; consumes one sequence number if no data
Half-close	Stop sending but keep receiving (e.g. sort-then-reply)

Exam lines

- TCP is virtually connection-oriented on top of unaware, connectionless IP (retransmission and reordering are TCP's job).
- Full-duplex use needs a three-way handshake: SYN (ISN, one seq), SYN+ACK (reverse ISN + rwnd), ACK (usually no seq consumption).
- SYN flooding is DoS by fake source IPs filling TCB/timer resources; limits, filters, and cookies are the taught defenses.
- Data is bidirectional with piggybacked ACKs; PSH is for interactive immediacy (often ignored); URG only marks special bytes in order.
- Close is three-way FIN / FIN+ACK / ACK, or four-way half-close so one side can finish sending while the other still replies.